

EXPERIMENT- 4

1. # Product Inventory Data

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
# Bar Chart
```

```
barplot(quantity,
```

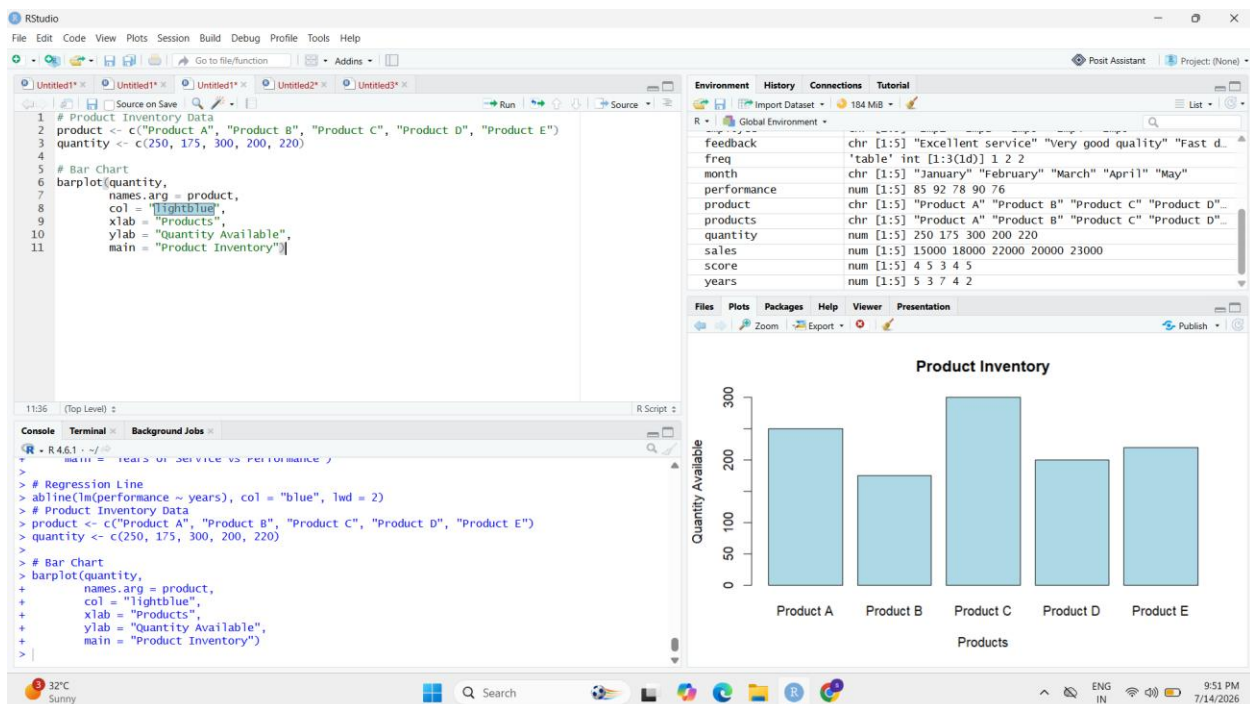
```
  names.arg = product,
```

```
  col = "lightblue",
```

```
  xlab = "Products",
```

```
  ylab = "Quantity Available",
```

```
  main = "Product Inventory")
```



2.# Product Category Data

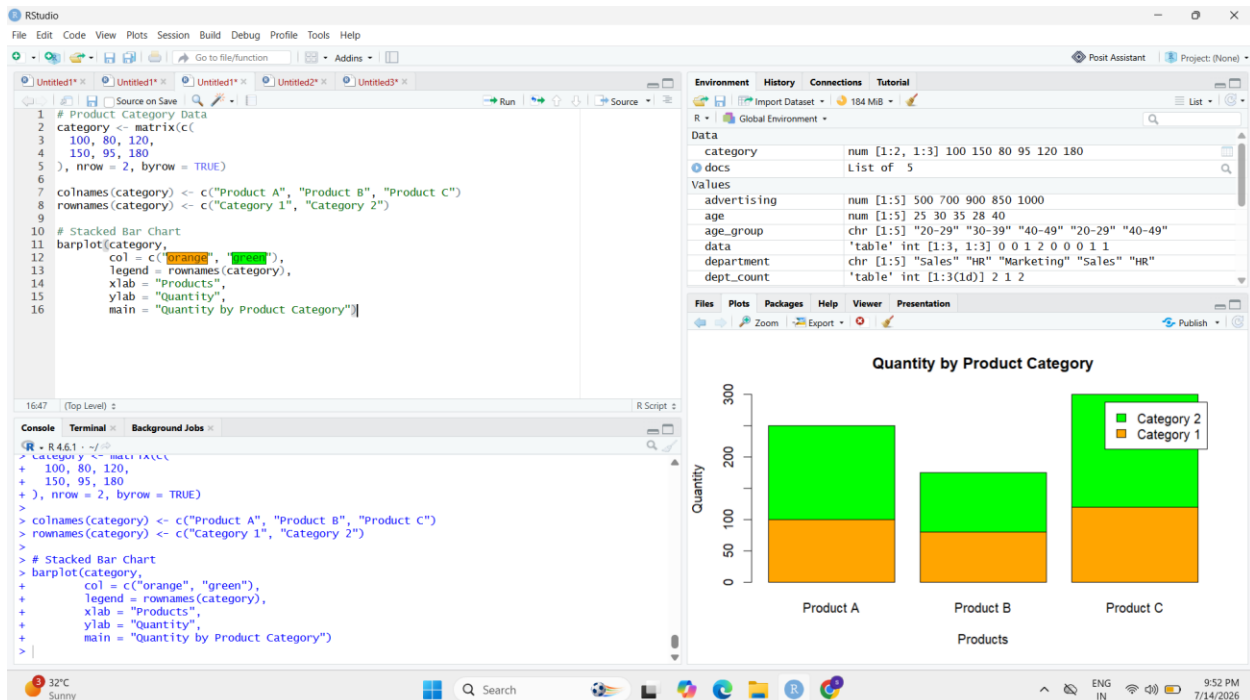
```
category <- matrix(c(
  100, 80, 120,
  150, 95, 180
), nrow = 2, byrow = TRUE)
```

```
colnames(category) <- c("Product A", "Product B", "Product C")
```

```
rownames(category) <- c("Category 1", "Category 2")
```

Stacked Bar Chart

```
barplot(category,
  col = c("orange", "green"),
  legend = rownames(category),
  xlab = "Products",
  ylab = "Quantity",
  main = "Quantity by Product Category")
```



3. # Product Data

```
price <- c(20, 15, 18, 25, 22)
```

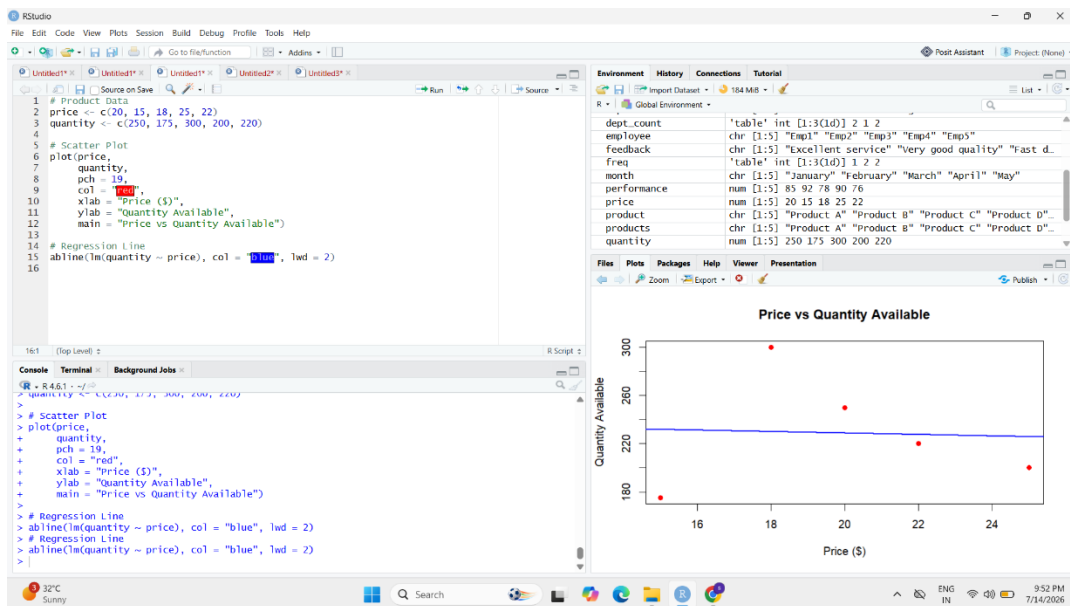
```
quantity <- c(250, 175, 300, 200, 220)
```

Scatter Plot

```
plot(price,  
      quantity,  
      pch = 19,  
      col = "red",  
      xlab = "Price ($)",  
      ylab = "Quantity Available",  
      main = "Price vs Quantity Available")
```

Regression Line

```
abline(lm(quantity ~ price),  
       col = "blue",  
       lwd = 2)
```



4.

